

We can now finish the proof of our theorem.

Proof of Theorem 1.2. Suppose on the contrary that $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. Since ζ is holomorphic at $1 + it_0$, it must vanish at least to order 1 at this point, hence

$$|\zeta(\sigma + it_0)|^4 \leq C(\sigma - 1)^4 \quad \text{as } \sigma \rightarrow 1,$$

for some constant $C > 0$. Also, we know that $s = 1$ is a simple pole for $\zeta(s)$, so that

$$|\zeta(\sigma)|^3 \leq C'(\sigma - 1)^{-3} \quad \text{as } \sigma \rightarrow 1,$$

for some constant $C' > 0$. Finally, since ζ is holomorphic at the points $\sigma + 2it_0$, the quantity $|\zeta(\sigma + 2it_0)|$ remains bounded as $\sigma \rightarrow 1$. Putting these facts together yields

$$|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \rightarrow 0 \quad \text{as } \sigma \rightarrow 1,$$

which contradicts Corollary 1.5, since the logarithm of real numbers between 0 and 1 is negative. This concludes the proof that ζ is zero free on the real line $\operatorname{Re}(s) = 1$.

1.1 Estimates for $1/\zeta(s)$

The proof of the prime number theorem relies on detailed manipulations of the zeta function near the line $\operatorname{Re}(s) = 1$; the basic object involved is the logarithmic derivative $\zeta'(s)/\zeta(s)$. For this reason, besides the non-vanishing of ζ on the line, we need to know about the growth of ζ' and $1/\zeta$. The former was dealt with in Proposition 2.7 of Chapter 6; we now treat the latter.

The proposition that follows is actually a quantitative version of Theorem 1.2.

Proposition 1.6 *For every $\epsilon > 0$, we have $1/|\zeta(s)| \leq c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \geq 1$, and $|t| \geq 1$.*

Proof. From our previous observations, we clearly have that

$$|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \geq 1, \quad \text{whenever } \sigma \geq 1.$$

Using the estimate for ζ in Proposition 2.7 of Chapter 6, we find that

$$|\zeta^4(\sigma + it)| \geq c|\zeta^{-3}(\sigma)| |t|^{-\epsilon} \geq c'(\sigma - 1)^3 |t|^{-\epsilon},$$

for all $\sigma \geq 1$ and $|t| \geq 1$. Thus

$$(3) \quad |\zeta(\sigma + it)| \geq c'(\sigma - 1)^{3/4}|t|^{-\epsilon/4}, \quad \text{whenever } \sigma \geq 1 \text{ and } |t| \geq 1.$$

We now consider two separate cases, depending on whether the inequality $\sigma - 1 \geq A|t|^{-5\epsilon}$ holds, for some appropriate constant A (whose value we choose later).

If this inequality does hold, then (3) immediately provides

$$|\zeta(\sigma + it)| \geq A'|t|^{-4\epsilon},$$

and it suffices to replace 4ϵ by ϵ to conclude the proof of the desired estimate, in this case.

If, however, $\sigma - 1 < A|t|^{-5\epsilon}$, then we first select $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$. The triangle inequality then implies

$$|\zeta(\sigma + it)| \geq |\zeta(\sigma' + it)| - |\zeta(\sigma' + it) - \zeta(\sigma + it)|,$$

and an application of the mean value theorem, together with the estimates for the derivative of ζ obtained in the previous chapter, give

$$|\zeta(\sigma' + it) - \zeta(\sigma + it)| \leq c''|\sigma' - \sigma||t|^\epsilon \leq c''|\sigma' - 1||t|^\epsilon.$$

These observations, together with an application of (3) where we set $\sigma = \sigma'$, show that

$$|\zeta(\sigma + it)| \geq c'(\sigma' - 1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma' - 1)|t|^\epsilon.$$

Now choose $A = (c'/(2c''))^4$, and recall that $\sigma' - 1 = A|t|^{-5\epsilon}$. This gives precisely

$$c'(\sigma' - 1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma' - 1)|t|^\epsilon,$$

and therefore

$$|\zeta(\sigma + it)| \geq A''|t|^{-4\epsilon}.$$

On replacing 4ϵ by ϵ , the desired inequality is established, and the proof of the proposition is complete.

2 Reduction to the functions ψ and ψ_1

In his study of primes, Tchebychev introduced an auxiliary function whose behavior is to a large extent equivalent to the asymptotic distribution of primes, but which is easier to manipulate than $\pi(x)$. **Tchebychev's ψ -function** is defined by

$$\psi(x) = \sum_{p^m \leq x} \log p.$$